

Kartik Dhawan

AI Developer

Skills

- Python
- Rest API
- Data Mining
- Pandas
- Flask / FastAPI
- Model Training & Evaluation
- Django
- SQL
- Scikit-learn
- PyTorch
- Predictive Analytics
- Machine Learning
- JavaScript
- NumPy
- Deep Learning
- Statistical Analysis

Professional Experience

AI Developer
Developer
Associate Developer

June 2025 - Present
April 2023 - May 2025
February 2021 - March 2023

Education

Bachelor of Technology - BTech,
Computer Science
Anand International College of
Engineering, Jaipur

Projects

Manufacturing Process Optimization

- Our client, a large manufacturing firm, required an AI-driven solution to optimize their manufacturing processes to improve efficiency, reduce operational costs, and enable predictive decision-making. The goal was to design intelligent models capable of identifying inefficiencies, predicting production outcomes, and recommending optimal process parameters using advanced Machine Learning and Data Science techniques.
- Utilized advanced data mining, statistical analysis, and Machine Learning techniques to identify bottlenecks and inefficiencies in manufacturing workflows, enabling data-driven process optimization.
- Designed and developed end-to-end data pipelines and predictive models using Python libraries such as NumPy, Pandas, Scikit-learn, and TensorFlow to analyze production data, forecast process outcomes, and support intelligent decision-making.
- Extracted, transformed, and processed large-scale production data from enterprise data systems using SQL, writing complex queries to aggregate, normalize, and prepare datasets for Machine Learning model training and evaluation.
- Designed and implemented analytical dashboards and performance reports to visualize key production metrics, model predictions, and efficiency trends, enabling stakeholders to monitor system performance and optimization results.

- Applied strong mathematical and statistical concepts including probability, regression, optimization techniques, and predictive modeling to design custom algorithms for process improvement rather than relying on pre-built AI APIs.
- Collaborated closely with production, engineering, and IT teams to understand system architecture, operational constraints, and business requirements, ensuring that AI/ML solutions aligned with real-world manufacturing environments.
- Developed simulation and predictive models to evaluate the impact of process changes, optimize resource allocation, and improve scheduling efficiency, ensuring scalable and production-ready AI solutions.
- Implemented intelligent optimization strategies such as dynamic scheduling, resource utilization modeling, and parameter tuning using Machine Learning models to improve overall manufacturing performance.

Demand Forecasting and Inventory Optimization using AI

- Developed an AI-driven demand forecasting and inventory optimization system for a retail client to improve supply chain efficiency, reduce stock shortages, and minimize excess inventory using predictive analytics and Machine Learning models.
- Collected historical sales, seasonal trends, and supply chain data from enterprise databases using SQL and Python for predictive modeling.
- Applied time-series forecasting techniques on historical sales data to generate predictive insights and support inventory planning.
- Implemented Machine Learning algorithms using Python libraries such as Pandas, NumPy, Scikit-learn, and TensorFlow to analyze sales patterns and customer behavior.
- Designed optimization algorithms to recommend optimal inventory levels and reorder points based on predicted demand and lead time.
- Built automated data pipelines for continuous model training and updating using new incoming data.
- Applied statistical analysis and probability theory to evaluate uncertainty and improve forecast reliability.
- Created dashboards and analytical reports to visualize demand trends, prediction results, and inventory performance metrics.
- Collaborated with business and supply chain teams to integrate AI models into operational decision-making processes.

Online Restaurant Reservation Platform Development

- Our client, a USA-based F&B firm, required an intelligent online reservation platform to streamline table booking operations while improving customer experience using data-driven insights and automated scheduling. The objective was to design and build a scalable system with predictive capabilities, optimized reservation handling, and intelligent resource allocation using Python-based backend architecture and analytical models.
- Leveraged Python, Django, and modern frontend technologies to design and develop a scalable reservation platform with a strong backend architecture, ensuring high performance, data consistency, and efficient handling of concurrent booking requests.
- Developed a feature-rich system that allowed users to browse available tables, select desired dates and times, and make reservations seamlessly, while implementing logic-based scheduling and availability validation to prevent conflicts and optimize table utilization.
- Designed customized user interfaces for both restaurant managers and customers, ensuring usability, efficient workflow management, and clear visualization of reservation data and system status.

- Implemented automated reservation confirmations via email and SMS, along with backend automation logic to handle booking validation, cancellation rules, and scheduling updates, reducing manual effort and improving operational efficiency.
- Developed a centralized dashboard for restaurant managers to monitor reservations, customer activity, and booking trends, and used data analysis techniques to generate insights for better decision-making and resource planning.
- Integrated payment processing using Stripe API to support secure prepayments and booking guarantees, ensuring reliable transaction handling and seamless user experience.
- Designed data models and backend services to store and process reservation data, enabling future use of Machine Learning models for demand prediction, peak-time analysis, and intelligent scheduling optimization.
- Applied analytical and statistical methods on booking data to identify usage patterns, improve table allocation strategies, and support the development of predictive features for reservation management.

Customer Behavior Analysis using Deep Learning

- Built an AI-driven customer behavior analysis system to understand user patterns, predict customer actions, and improve business decision-making using Deep Learning and advanced data analytics.
- Extracted and processed large customer datasets from multiple sources using Python, SQL, and data preprocessing techniques.
- Used advanced analytics and machine learning techniques to analyze customer behavior patterns and generate predictive insights.
- Applied clustering, classification, and regression algorithms to segment customers and predict future actions.
- Used feature engineering, normalization, and dimensionality reduction techniques to improve model performance.
- Designed data pipelines for training, validation, and deployment of Machine Learning models.
- Built analytical dashboards to visualize predictions, customer segments, and performance metrics.
- Worked with business teams to integrate model outputs into marketing and recommendation strategies.

AI-Based Recommendation Engine with Custom Machine Learning Models

- Developed a scalable AI-based recommendation engine for a large e-commerce platform to provide personalized product recommendations using custom Machine Learning algorithms. The objective was to design intelligent models capable of learning user behavior, predicting preferences, and optimizing recommendations without relying on third-party AI APIs.
- Collected and processed large-scale user interaction data, including clicks, purchases, search history, and browsing patterns using Python and SQL to build structured datasets for Machine Learning training.
- Implemented recommendation algorithms using similarity analysis and behavioral data to generate personalized suggestions.
- Developed custom Machine Learning models to predict user preferences using classification, regression, and similarity-based techniques.
- Built data pipelines for data preprocessing, feature engineering, model training, and evaluation to support continuous learning.
- Designed backend architecture to support real-time recommendation generation and scalable deployment.

- Collaborated with product and engineering teams to integrate the recommendation engine into the production system and align results with business objectives.